 # Operational Memo -- Scenario C: Governance Across 14, 6, and 0 Crew Regimes

****To:** SDV Program Office & Vessel Operations Command**
From: Principal Systems Architect, ‘sdv-resource-governor‘
Subject: Deterministic Resource, Deck-Safety, and Autonomy Rules Under Crew Regimes **14**, **6**, and **0**

1. Executive Summary
A Software-Defined Vessel (SDV) cannot treat human crew availability as a mere scheduling annotation: crew count directly governs **which physical payload transitions are admissible**, **which manual deck evolutions can be staffed**, **whether at-sea vehicle replacement (A’ window) is physically safe**, and **how fault recovery escalates when telemetry degrades or a payload overconsumes**. In ‘sdv-resource-governor‘, human crew availability, flight-deck clearance (‘deck_clear‘), and port-only installation constraints are enforced as hard operational guards inside ‘scheduling.py‘, ‘admission.py‘, and ‘replay.py‘.

2. Quantitative Regime Comparison (‘FULL_CREW_14‘ vs. ‘REDUCED_CREW_6‘ vs. ‘UNCREWED_0‘)

Operational Dimension	Regime 1: 14 Crew (FULL_CREW_14)	Regime 2: 6 Crew (REDUCED_CREW_6)	Regime 3: 0 Crew (UNCREWED_0)
Bridge & Engine Watch	5 personnel	4 personnel	0 (Remote Shore Control + Onboard Autonomy)
Available Deck/Payload Crew	9 personnel	2 personnel	0 personnel
Max Concurrent Manual Deck Teams	2 teams (for non-conflicting manual tasks)	1 team (strictly serialized)	0 manual deck evolutions
G4 UAV Launch / Recovery (5-12 kn into wind, wind <= 25 kn, SS <= 4, deck_clear=true)	PERMITTED (2 deck crew; 15-60 s local bridge confirmation)	PERMITTED (2 deck crew; 60-180 s watch handover confirmation)	FORBIDDEN (requires manual flight-deck handling; only stowed/standby modes)
G2 Winch / Probe Deployment (0-5 kn, wind <= 20 kn, SS <= 4, deck_clear=true)	PERMITTED (2 deck crew; cannot overlap G4 flight ops)	PERMITTED (2 deck crew; serialized)	FORBIDDEN (requires umbilical/winch deck handling)
Simultaneous G2 Winch + G4 Flight Ops	FORBIDDEN (Annex A.5 requires flight deck clear during G4 ops; speed windows 0-5 kn vs 5-12 kn into wind are disjoint)	FORBIDDEN (insufficient crew + Annex A.5 deck clearance + disjoint speed windows)	FORBIDDEN (0 crew)
At-Sea Vehicle Replacement A’ (4-8 kn, wind <= 20 kn, SS <= 3; G4: 3 crew/30m, G3: 4 crew/45m)	PERMITTED (available_crew=9 >= 4 for G3 and >= 3 for G4)	BLOCKED (available_crew=2 < 3 for G4 and < 4 for G3 safe replacement at sea)	FORBIDDEN (0 crew; all vehicle replacements deferred to port)
At-Sea Bay/Mast or >250 kg Module Install (G5, G6 420 kg in bay2/bay1_a)	FORBIDDEN AT SEA (Annex A.5 bay/mast installs are port-only; Annex A.4 limits at-sea packages to <= 250 kg)	FORBIDDEN AT SEA (port-only bay/mast + >250 kg + available_crew=2 < 4)	FORBIDDEN AT SEA (CRW-02: zero physical installations at sea)
Autonomous Sensor & Compute Guests (G1, G3 in-water survey, G5, G6 auto-sampler)	100% Operational	100% Operational	100% Operational (governed via software mode step-down, throttle, and power shed)

---

3. Concrete Operational Rules & Physical Interlocks Enforced by the Solver

1. **Why G2 and G4 Cannot Execute Simultaneous Deck Operations Even With 14 Crew (‘Annex A.5‘)**:
   - While **14 crew** provides ‘9‘ deck/payload personnel (enough to staff two manual handling teams for non-conflicting tasks), **G2 ROV/winch deployment and G4 UAV launch/recovery are physically incompatible**: (a) Annex A.5 mandates that *"Launch or recovery of a drone requires the flight deck to be clear of other ongoing deck operations"* (‘state.deck_clear == True‘), and (b) **G2** requires low-speed station-keeping (‘0-5 kn‘) whereas **G4** requires ‘5-12 kn‘ with vessel heading aligned within ‘+/- 15 deg‘ of the wind.
2. **Why G6 (‘420 kg‘, ‘bay2‘/‘bay1_a‘) Is Strictly Port-Only Across All Crew Regimes (‘Annex A.4 & A.5‘)**:
   - Annex A.5 specifies that payload installation in **Bays 1-2** and **Mast Positions 1-3** is a **port-only operation**, and Annex A.4 restricts at-sea A’ package installation to light deck packages **‘<= 250 kg‘** in ‘Sea State <= 3‘, ‘wind <= 20 kn‘, and ‘4-8 kn‘. Because **G6** weighs ‘420 kg > 250 kg‘ and installs into ‘bay2‘ / ‘bay1_a‘, ‘G6‘ has ‘port_only: true‘ and is rejected (‘NO‘ / ‘NOT_NOW‘) for at-sea installation under **14**, **6**, and **0** crew alike.
3. **Acoustic Survey Deck-Machinery Lockout (‘G3‘ ‘forbids_deck_machinery_while_active=True‘)**:
   - Across both **14** and **6** crew regimes, while **G3** is in ‘SURVEY_QUIET‘ (‘1 kW‘, ‘2-6 kn‘), hydraulic A-frames, winches (**G2**), and flight-deck handling gear (**G4**) are locked out to prevent acoustic corruption of the seabed survey sonar.
   - Before turning into the wind or operating deck machinery for **G4** recovery, the governor must issue ‘ACOUSTIC_HOLD‘ to **G3**, verify a causal ‘ACOUSTIC_HOLD_CONFIRMED‘ ACK from ‘guest_id=’G3’‘ matching the active ‘command_id‘ and ‘plan_version‘ (‘15 s‘ nominal acoustic handshake, ‘60 s‘ timeout), and confirm **G3** has transitioned to ‘HOLD_ACOUSTIC‘.
4. **Separation of Normal Flight Window vs. A’ At-Sea Replacement Window (‘CRW-03‘)**:
   - **Normal G4 Flight Launch/Recovery**: vessel speed **‘5-12 kn‘**, wind **‘<= 25 kn‘**, Sea State **‘<= 4‘**, heading aligned within ‘+/- 15 deg‘ of wind, ‘deck_clear=True‘, **‘2‘ deck crew** (permitted under **14** and **6** crew).
   - **A’ At-Sea Vehicle Replacement**: vessel speed **‘4-8 kn‘**, wind **‘<= 20 kn‘**, Sea State **‘<= 3‘**, ‘deck_clear=True‘, **‘3‘ deck crew / ‘30 min‘** for **G4** or **‘4‘ deck crew / ‘45 min‘** for **G3** (permitted only under **14** crew; blocked under **6** and **0** crew).

---

4. Fault Governance & Decision Escalation Under Reduced (6) and Uncrewed (0) Regimes

When operating with **6** or **0** crew, manual intervention on tripped breakers, overconsuming payloads, or degraded cooling loops is delayed (**6 crew**: ‘60-180 s‘ watch handover) or impossible (**0 crew**). The governor enforces three automated fail-safe barriers:

- **Urgent Safety Commands (‘STOP‘, ‘HOLD‘, ‘SECURE <= 200 ms‘, ‘SAF-01‘)**: Routed strictly over the dual ‘10 Gbps‘ local shipboard LAN (‘< 5 ms‘ transport latency), never over ‘600-800 ms‘ SATCOM. Measured execution latency on both MacBook Pro (‘arm64‘) and the live Arduino Ventuno Q (‘training06‘, ‘aarch64‘) is **‘< 1 ms‘** (‘<< 200 ms‘ deadline).
- **Guest Overconsumption (‘+10%‘ to ‘+20%‘, ‘RUN-04‘)**: Because none of ‘G1‘-‘G5‘ is safety-critical by design, if a payload (e.g., **G5** declared ‘45 kW‘ drawing ‘54 kW‘ = ‘+20%‘) exceeds its manifest envelope, 1:1 thermal coupling immediately adds ‘+9 kW‘ to the chilled-water ledger and the governor autonomously throttles **G5** to ‘DEGRADED_70PCT_LOSS‘ (‘8 kW‘) while protecting active uninterruptible/recovery tasks (**G3** recharge, **G4** flight/recovery).
- **Lost or Stale Telemetry Feeds (‘RUN-03‘)**: If ‘seawater_temp_c‘ exceeds its TTL (‘STALE‘ / ‘UNAVAILABLE‘), the governor never extrapolates optimistic cooling; it clamps to the conservative **‘30 deg C‘ (‘120 kW‘ nominal / ‘72 kW‘ degraded)** thermal bound until sensor freshness is restored.